Tests for chromosomal abnormalities

Optional routine screening tests in the first trimester assess the risk of your baby having a chromosomal disorder such as Down syndrome. If your risk is high, you will be offered a diagnostic test for a definite result.

What is screened for

In addition to Down syndrome, screening tests also assess the risk of other chromosomal abnormalities such as Patau syndrome (trisomy 13) and Edwards syndrome (trisomy 18). Babies with these conditions have more severe mental and physical abnormalities than babies with Down syndrome and rarely survive beyond a year. These conditions are rare, with each year about 1 in 10,000 babies born with trisomy 13 and 1 in 6,000 born with trisomy 18, compared to 1 in 800 babies born with Down syndrome.

The "combined" test

The recommended screening test for Down, and the one most commonly offered, is the "nuchal translucency screening," performed between 11 and 14 weeks. This test has a high accuracy rate and the results are produced quickly.

The test uses a combination of a maternal blood test and an ultrasound of your baby. The ultrasound measures the thickness of the skin at the back of your baby's neck, called the nuchal fold (see Nuchal translucency screening). The blood test looks at the levels of two chemicals: pregnancy associated plasma protein A (PAPP-A) and one of the pregnancy hormones, human chorionic gonadotropin (hCG). The results of the blood test are combined with your age and the measurement of the nuchal fold. A mathematical formula is then used to calculate your baby's risk of Down syndrome.

If you have your blood test before your ultrasound, you will usually receive your results immediately following your ultrasound. If you have your blood drawn at the time of the ultrasound, you will receive your results a few days later. Your baby's risk of Down, Edwards, and Patau syndrome from the combined test may be higher, lower, or the same as her risk based on your age alone.

Noninvasive testing

DNA is the genetic material we get from our parents that determines characteristics such as eye color and any genetic diseases. During pregnancy free DNA from the baby is present in the maternal circulation. When blood from the mother is analyzed for the presence of chromosome 21, 18, and 13, higher than average amounts predicts that the baby may have one of these syndromes. This test is the most accurate screening test available for the detection of Down, Edwards, and Patau syndrome, with detection rates